

A Lightweight Edge-Centric Chewing Sensing System with Multimodal Biosignal Fusion for Oral Health Monitoring

Yuanhao Feng
The University of
Electro-Communications
Tokyo, Japan
fengyuanhaohaohao@gmail.com

Youwei Zhang
The University of
Electro-Communications
Tokyo, Japan
zhanguv@uec.ac.jp

Hao Zhou
University of Science and Technology
of China
Anhui, China
kitewind@ustc.edu.cn

Xiang Zhang*
University of Science and Technology
of China
Anhui, China
zhangxiang@ieee.org

Zhi Liu*
The University of
Electro-Communications
Tokyo, Japan
liu@ieee.org

Abstract

Tooth health is essential to overall well-being, yet remains largely unmonitored outside of dental clinics. We present **NeuroChew**, a multimodal wearable system that infers oral health status from everyday chewing behavior. By integrating electromyography (EMG), electrooculography (EOG), and electroencephalography (EEG), NeuroChew captures neuromuscular, attentional, and cognitive signals related to chewing. From these signals, the system extracts features such as symmetry, rhythm regularity, and EEG-based cognitive strain, enabling classification into three health-related chewing states: Normal, Asymmetric, and Discomfort. A lightweight Transformer-based model performs the classification, augmented by a rule-based refinement layer to improve interpretability and robustness. In a 20-participant pilot study, NeuroChew achieved 83.7% accuracy, a macro F1 score of 0.826, and a macro AUC of 0.893 in distinguishing the three chewing states. These results demonstrate the feasibility of behavior-based, non-invasive oral health monitoring using wearable biosignals, and point toward new opportunities for ubiquitous dental diagnostics in everyday life. The system is designed with real-time processing in mind, supporting future deployment across edge-cloud infrastructures and autonomous agents.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing design and evaluation methods; • **Hardware** → Signal processing systems; Sensor applications and deployments.

Keywords

tooth health, chewing behavior, wearable sensing, multimodal biosignals, oral health monitoring

*Co-corresponding author.



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1 Introduction

Despite the critical role of tooth health in nutrition, speech, and quality of life, its assessment remains confined to occasional clinic visits [20]. Subtle symptoms—such as chewing asymmetry, discomfort, or force suppression—often go undetected in daily life [26]. This gap highlights the need for continuous, non-invasive monitoring. As chewing is both frequent and functionally linked to oral health, it provides a natural behavioral window for sensing tooth-related conditions in situ [12].

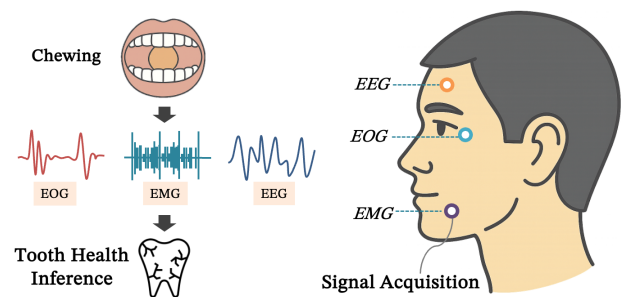


Figure 1: Demonstration of NeuroChew system. Chewing behavior is captured through multimodal biosignals including EMG, EOG, and EEG. Extracted features such as symmetry, irregularity, and cognitive strain are used to infer tooth health status.

Chewing is primarily driven by coordinated contractions of the masseter and temporalis muscles, which generate surface EMG signals that reflect bite force and symmetry. In healthy individuals, chewing alternates between both sides of the mouth with consistent rhythm and amplitude. However, tooth discomfort—caused by cavities, malocclusion, or localized sensitivity—often leads to

asymmetric muscle activation, reduced chewing effort, or avoidance behaviors. These neuromuscular changes are readily observable in EMG patterns. Moreover, discomfort may co-occur with altered attentional or cognitive states, which can be captured through EOG and EEG signals during chewing tasks.

Designing such a system presents several challenges. **First**, chewing must be reliably distinguished from other facial activities like talking or swallowing, which share overlapping EMG patterns [2]. **Second**, there is no direct mapping from behavior to specific dental conditions—pain or asymmetry must be inferred through indirect signals. **Third**, chewing behavior varies widely across individuals due to habits, dental anatomy, and dietary preferences. **To address these**, we adopt a multimodal sensing approach combining EMG, EOG, and EEG signals to capture muscular activation, attentional patterns, and cognitive strain. A feature extraction pipeline transforms these into behavioral indicators such as bilateral symmetry, rhythm regularity, and discomfort-related neural responses. We apply a lightweight classifier enhanced with rule-based postprocessing and a personalization module to accommodate user-specific baselines.

Fig. 1 demonstrates NeuroChew’s multimodal sensing pipeline for tooth health inference. In summary, our contributions are as follows:

- We design and implement **NeuroChew**, the first multimodal wearable system that infers oral health status from chewing behavior using EMG, EOG, and EEG signals.
- We propose a lightweight Transformer-based inference pipeline augmented with rule-based heuristics, enabling interpretable classification into Normal, Asymmetric, and Discomfort chewing states.
- We validate NeuroChew through a 20-participant pilot study, achieving 83.7% accuracy and 0.826 macro F1 across three chewing classes, demonstrating the feasibility of behavior-based oral health sensing.

2 Related Work

Neuromuscular and Attentional Sensing via EMG, EOG, and EEG. A growing body of research explores the use of bioelectrical signals—including EMG, EOG, and EEG—for inferring human physiological and cognitive states [3–5, 10, 15, 16, 22, 23, 25, 27, 32, 36, 37]. EMGSense [10] infers respiration and physical activity from subtle EMG fluctuations, while sEMG-Gesture [37] performs fine-grained hand gesture recognition using adaptive feature learning. WINCE [25] repurposes wearable EOG glasses to detect facial action units for unobtrusive pain and affect sensing. In EEG-based systems, DeepBrain [32] enables intention-driven robot control using low-cost brain signals, and AreWeInTheZone [36] detects shared cognitive flow between team members via EEG synchrony. These systems highlight the potential of wearable biosignals for real-time inference of user state and intent. *NeuroChew* extends this line of work by integrating EMG, EOG, and EEG to sense neuromuscular coordination and cognitive strain during chewing, enabling non-invasive inference of oral health rather than task execution or emotion classification.

Oral health sensing. Recent advances in wearable and ambient computing have enabled new approaches to oral health sensing and

behavior inference [1, 6, 8, 11, 12, 14, 17, 19, 24, 31, 33, 34, 38]. Recent systems have investigated oral health sensing using acoustic, inertial, and embedded sensor modalities. Chewpin [6] detects chewing and swallowing via collar-mounted microphones. IOTeeth [12] embeds piezoelectric sensors in dental appliances to monitor occlusion patterns and identify bite disorders. mTeeth [1] uses wrist-worn inertial sensors to classify brushing surfaces and transitions. ToMoBrush [34] analyzes toothbrush acoustic resonances to detect food debris and dental anomalies. ToothFairy [31] leverages reversed earphone signals to localize brushing actions through signal attenuation without machine learning. While these systems achieve high accuracy in controlled scenarios, they primarily focus on brushing activity or sound-based detection, and generally lack insight into discomfort, effort, or neuromuscular asymmetry. In contrast, *NeuroChew* targets natural chewing as a rich physiological signal source. This design enables non-invasive, behavior-driven inference of oral health in everyday settings.

Multimodal biosignal fusion. Multimodal fusion has emerged as a key strategy for enhancing the robustness of mobile health and behavior sensing systems [7, 9, 13, 18, 21, 28, 29, 35]. *HabitSense* [11] integrates EMG, EEG, and PPG within an AI-enhanced wearable platform to enable privacy-preserving mobile health applications. *Hide-and-Seek* [21] fuses facial EMG, ECG, and GSR to estimate emotional workload in service contexts. *Ubi-SleepNet* [35] focuses on sleep stage classification using motion and cardiac signals, comparing early, late, and hybrid fusion strategies, including attention and bilinear pooling. *M3BAT* [18] proposes a multi-branch adversarial training approach for domain adaptation across modalities and user populations. While these systems advance the state of multimodal learning in mobile and health settings, *NeuroChew* is distinct in applying biosignal fusion (EMG, EOG, EEG) to natural chewing behavior. It demonstrates how lightweight Transformer models combined with interpretable features can support real-time, non-invasive inference of oral health states in everyday environments.

3 System Design

NeuroChew is structured as a modular sensing pipeline that transforms raw chewing-related biosignals into actionable health insights. It comprises four key components: (1) signal acquisition and preprocessing, (2) feature extraction, (3) tooth health inference, and (4) personalization and user feedback. The overall architecture is shown in Figure 2.

Signal Acquisition and Preprocessing. To reliably distinguish chewing from similar facial activities such as speaking or blinking, *NeuroChew* employs a multimodal sensing setup with synchronized EMG, EOG, and EEG electrodes. EMG sensors on the bilateral masseter and temporalis muscles capture jaw activation patterns, while EOG electrodes near the outer canthi detect eye motion to contextualize chewing intent. EEG electrodes on the frontal scalp record neural responses associated with attention and discomfort. All signals are sampled at 512 Hz. To minimize noise and isolate chewing segments, we apply modality-specific filtering (e.g., high-pass for EMG, bandpass for EEG/EOG) and detect chewing episodes via energy thresholding and peak analysis. Segmenting only high-confidence chewing windows (2–3 seconds) ensures that downstream processing focuses on functionally relevant behavior.

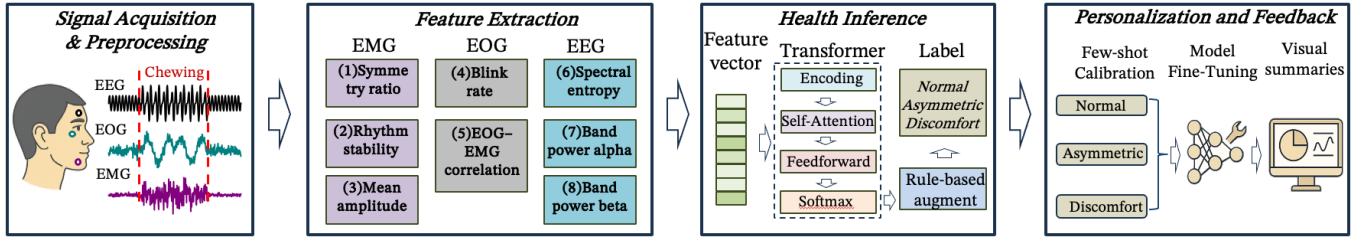


Figure 2: System overview of NeuroChew.

Feature Extraction. Since tooth health cannot be directly observed from biosignals, NeuroChew extracts a compact, interpretable set of features from each modality to infer latent oral states. From EMG, we compute (1) symmetry ratio (left/right RMS) to capture bilateral muscle coordination, (2) rhythm stability (inter-burst interval variance) to quantify regularity of chewing cycles, and (3) mean amplitude as a proxy for chewing effort. From EOG, we extract (4) blink rate, which reflects attentional state and cognitive strain, and (5) EOG–EMG rhythm correlation to assess motor-attentional synchrony. EEG features include (6) spectral entropy, indicating neural complexity and discomfort-related cognitive load, and (7–8) alpha and beta power, which reflect relaxed and alert cortical activity respectively.

These features exhibit distinct profiles across chewing states: Normal segments show high EMG symmetry, stable blink-EMG synchrony, and low entropy; Asymmetric segments present lateral imbalance and mild desynchronization; and Discomfort segments reveal reduced amplitude, erratic rhythm, and elevated EEG entropy. All eight features are concatenated into a unified vector $\mathbf{f}_i \in \mathbb{R}^8$, enabling the downstream classifier to robustly infer health states from multimodal chewing behavior with minimal computational overhead.

Health Inference. To bridge behavioral signals and latent health states, we formulate chewing classification as a three-way supervised task: Normal, Asymmetric, and Discomfort. A lightweight Transformer-based classifier [30] models temporal and multimodal dependencies, combining self-attention with a softmax output layer. It is trained with class-balanced cross-entropy loss and outputs confidence scores per segment.

To address model interpretability and low-confidence predictions, we incorporate a rule-based augmentation module that checks three handcrafted signal conditions: (1) persistent unilateral dominance, (2) amplitude suppression, and (3) rhythm disruption. When aligned with ambiguous predictions, these rules bias the decision toward the matching health state. This hybrid design improves transparency and robustness, allowing the system to flag discomfort even in early-stage or borderline chewing patterns, and reduces reliance on fully annotated data.

Personalization and Feedback. To adapt to inter-individual variability, NeuroChew supports lightweight personalization and longitudinal feedback. During an optional calibration phase, users complete a few labeled chewing trials, which are used to fine-tune the final classification layer via few-shot adaptation. This enables the model to account for individual differences in chewing strength, rhythm, and asymmetry. User-specific thresholds are also computed to support rule-based adjustments, such as amplitude suppression

or lateral dominance. After inference, NeuroChew presents visual feedback summarizing chewing balance, discomfort prevalence, and segment quality. Trends are tracked over time, enabling detection of behavioral drift or emerging discomfort. When threshold-crossing patterns are observed, the system can recommend recalibration or alert users to seek dental attention. All data is stored locally, supporting privacy-preserving deployment. This module bridges real-time inference with long-term oral health support, enhancing both usability and personalization.

4 Evaluation

4.1 Experimental Setup

Data Collection Setup. We collect EMG, EOG, and EEG signals using 6 adhesive electrodes placed on the face (Fig. 3). EMG sensors are placed over masseter and temporalis muscles, EOG near the outer canthi, and EEG on the forehead. All signals are sampled at 512 Hz using a synchronized acquisition board, enabling multimodal chewing behavior monitoring with minimal interference. This setup is designed to support future deployment across edge–cloud infrastructures with lightweight, real-time signal processing.

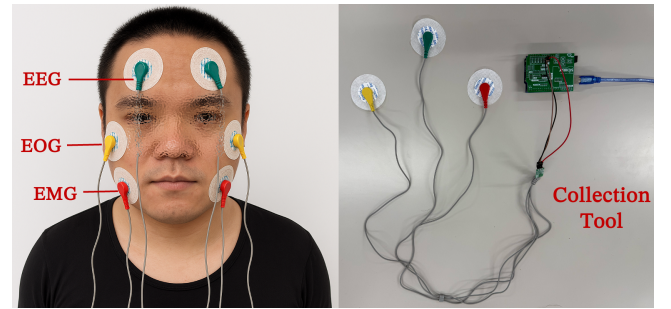


Figure 3: Data collection setup.

Participant Demographics and Setup. We conduct a user study with 20 participants (10 male, 10 female, ages 22–37). Four participants report diagnosed dental issues (e.g., cavities, missing molars), while the others are asymptomatic. All participants provide informed consent and follow a structured chewing protocol under lab supervision.

Chewing Tasks. Each participant performs three types of chewing sessions:

- (1) *Bilateral chewing:* Chewing normally using both sides of the mouth alternately.

- (2) *Unilateral chewing*: Chewing restricted to either the left or right side.
- (3) *Simulated discomfort*: Reduced-force chewing or jaw clenching to mimic oral pain or avoidance behavior.

Each task lasts approximately 2 minutes and is repeated twice with rest intervals. Multimodal signals are recorded using synchronized wearable sensors as described in Section 2.

Labeling and Annotation. A total of 360 chewing segments (approximately 18 per participant) are labeled into three categories: Normal, Asymmetric, and Discomfort. Labels are assigned based on the task condition, self-reported experience post-task, and manual video/signal inspection by two trained raters. The Asymmetric class captures habitual or task-induced lateral bias without reported pain; Discomfort segments reflects low-amplitude, irregular chewing patterns typically reported as uncomfortable.

4.2 Results

Table 1: Performance of NeuroChew on three-class classification (mean $\pm$ std across folds).

Metric	Accuracy	Macro F1	ROC AUC (macro)
NeuroChew	83.7% $\pm$ 3.1	0.826	0.893

Classification Performance. We evaluated the Transformer-based classifier using subject-independent 5-fold cross-validation. Performance metrics are summarized in Tab. 1. NeuroChew achieves an average accuracy of 83.7%, a macro F1 score of 0.826, and a macro-averaged ROC AUC of 0.893, demonstrating strong three-class discrimination across Normal, Asymmetric, and Discomfort chewing states. These results indicate that NeuroChew effectively captures multimodal differences in chewing behavior, enabling accurate detection of both motor-level asymmetry and discomfort-related neural patterns. The classifier maintains balanced performance across all three classes and shows robustness to cross-subject variability.

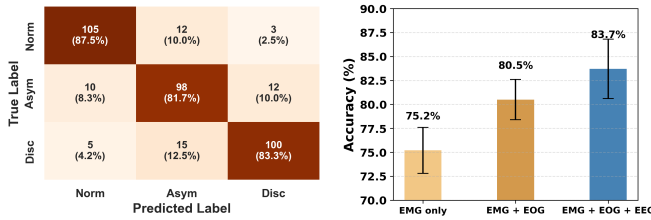


Figure 4: Evaluation results.

Fig. 4a shows the confusion matrix for the three-class chewing classification task. The classifier achieves strong inter-class separability, particularly in identifying Normal segments with high precision and recall. Most misclassifications occur between Asymmetric and Discomfort, which is expected due to their overlapping behavioral signatures—e.g., lateral chewing with reduced amplitude may reflect either a habitual asymmetry or emerging discomfort. This

confusion highlights the subtlety of chewing-related health inference and motivates the integration of rule-based cues to improve robustness in borderline cases.

Ablation Analysis. To assess the contribution of each sensing modality, we conducted an ablation study by removing one or more signal sources from the input. As shown in Fig. 4b, using EMG alone yields 75.2% accuracy, while combining EMG with EOG improves performance to 80.5%. Full fusion of EMG, EOG, and EEG achieves the best result at 83.7%, confirming that each modality provides complementary information. Notably, EEG adds cognitive context that helps distinguish subtle discomfort from asymmetry, while EOG improves attentional interpretation and noise robustness. These results validate our multimodal design and demonstrate that chewing health inference benefits from both muscular and neuro-cognitive signals.

User-Level Observations. For the four participants with known dental conditions, NeuroChew correctly identified 16 out of 18 relevant segments as either Asymmetric or Discomfort. Among healthy participants, some false predictions occurred during low-effort chewing or ambiguous rhythm. These findings highlight the need for personalized baselines and context-aware calibration in future deployments.

5 Discussion

NeuroChew introduces a novel oral health monitoring approach using chewing behavior as a multimodal sensing window. Our results show that variations in symmetry, rhythm, and effort can be captured via EMG, EOG, and EEG signals, enabling detection of motor and cognitive cues linked to discomfort or compensation. While the pilot study demonstrates feasibility, the dataset is small, and generalization to unsupervised, real-world use requires broader populations, including elderly and dental patients. The current three-class scheme—Normal, Asymmetric, Discomfort—supports interpretability but lacks the nuance of continuous oral health. Future work may explore regression models or personalized discomfort scales. We also envision integration with mobile health platforms and clinical systems for real-time tracking, early alerts, and long-term interventions at home or in clinics.

6 Conclusion

NeuroChew demonstrates the feasibility of inferring oral health status through natural chewing behavior using multimodal biosignals. By combining EMG, EOG, and EEG features within a lightweight classification pipeline, the system accurately distinguishes Normal, Asymmetric, and Discomfort states. Rule-based refinement, personalization, and feedback enhance interpretability and user engagement. This work paves the way for non-invasive, behavior-driven dental monitoring with applications in home care, preventive diagnostics, and ubiquitous health sensing.

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